

PRGM: 06

server.js

```
const http = require("http");
```

```
const server = http.createServer(function(req, res) {
```

```
    res.writeHead(200, {
        "Content-Type": "text/html"
    });
```

```
    res.write(`
        <html>
        <head>
            <title>Node.js Server</title>
        </head>
        <body>
            <h1>Welcome to Node.js</h1>
            <p>This HTML page is sent by a Node.js server.</p>
        </body>
        </html>
    `);
```

```
    res.end();
```

```
});
```

```
server.listen(3000, function() {
```

```
    console.log("Server running at http://localhost:3000");
```

```
});
```

PRGM: 01

index.html

<!DOCTYPE html>

<html>

<head>

 <title>Literary Haven</title>

 <meta name="viewport"

 content="width=device-width, initial-scale=1.0">

 <link rel="stylesheet" href="style.css">

</head>

<body>

<header>

 <h1>Literary Haven</h1>

 <nav>

 Home

 Books

 Authors

 Contact

 </nav>

</header>

<main>

<section id="home">

 <h2>Welcome to Literary Haven</h2>

 <p>

Discover wonderful books and famous authors.

</p>

</section>

<section id="books">

<h2>Popular Books</h2>

<div class="books">

<article>

<h3>Pride and Prejudice</h3>

<p>Written by Jane Austen</p>

<button onclick="showBook('Pride and Prejudice')">

Read More

</button>

</article>

<article>

<h3>The Hobbit</h3>

<p>Written by J. R. R. Tolkien</p>

<button onclick="showBook('The Hobbit')">

Read More

</button>

</article>

<article>

<h3>Sherlock Holmes</h3>

<p>Written by Arthur Conan Doyle</p>

<button onclick="showBook('Sherlock Holmes')">

Read More

</button>

</article>

</div>

</section>

<section id="authors">

<h2>Famous Authors</h2>

Jane Austen

J. R. R. Tolkien

Arthur Conan Doyle

</section>

<section id="contact">

<h2>Contact Us</h2>

<form onsubmit="return contactMessage()">

<label>Name:</label>

<input type="text" id="name" required>

<label>Email:</label>

<input type="email" required>

<button type="submit">Submit</button>

</form>

```
</section>
```

```
</main>
```

```
<footer>
```

```
  <p>© 2026 Literary Haven</p>
```

```
</footer>
```

```
<script src="script.js"></script>
```

```
</body>
```

```
</html>
```

style.css

```
body {  
  font-family: Arial;  
  margin: 0;  
  background: #f2f2f2;  
}
```

```
header {  
  background: darkblue;  
  color: white;  
  text-align: center;  
  padding: 20px;  
}
```

```
nav a {  
  color: white;  
  text-decoration: none;  
  margin: 10px;
```

```
}
```

```
section {  
    padding: 30px;  
    text-align: center;  
}
```

```
img {  
    max-width: 100%;  
}
```

```
.books {  
    display: flex;  
    justify-content: center;  
    gap: 20px;  
    flex-wrap: wrap;  
}
```

```
article {  
    background: white;  
    padding: 20px;  
    width: 200px;  
    border: 1px solid gray;  
}
```

```
button {  
    padding: 8px 15px;  
    background: darkblue;  
    color: white;  
    border: none;  
}
```

```
li {  
    list-style-position: inside;  
    margin: 8px;  
}
```

```
form {
    display: flex;
    flex-direction: column;
    gap: 10px;
    max-width: 300px;
    margin: auto;
}

input {
    padding: 8px;
}

footer {
    background: darkblue;
    color: white;
    text-align: center;
    padding: 15px;
}

/* Responsive Design */

@media (max-width: 600px) {

    nav {
        display: flex;
        flex-direction: column;
        gap: 10px;
    }

    .books {
        flex-direction: column;
        align-items: center;
    }
}
```

```
    article {  
        width: 80%;  
    }  
}
```

script.js

```
function showBook(book) {  
    alert("You selected: " + book);  
}
```

```
function contactMessage() {  
    var name = document.getElementById("name").value;  
  
    alert("Thank you, " + name + "!");  
  
    return false;  
}
```

PRGM: 02

index.html

```
<!DOCTYPE html>
```

```
<html>
```

```
<head>
```

```
    <title>Registration Form</title>
```

```
    <link rel="stylesheet" href="style.css">
```

```
</head>
```

```
<body>
```

```
<h1>Student Registration</h1>
```

```
<form onsubmit="return validateForm()">
```

```
    <label>Name:</label>
```

```
    <input type="text" id="name">
```

```
    <label>Email:</label>
```

```
    <input type="email" id="email">
```

```
    <label>Phone:</label>
```

```
    <input type="text" id="phone">
```

```
    <label>Password:</label>
```

```
    <input type="password" id="password">
```

```
    <label>Gender:</label>
```

```
    <input type="radio" name="gender"> Male
```

```
    <input type="radio" name="gender"> Female
```

```
    <label>Course:</label>
```

```
    <select id="course">
```

```
        <option>B.Sc</option>
```

```
        <option>M.Sc</option>

        <option>BCA</option>

        <option>MCA</option>
    </select>

    <button type="submit">Register</button>

    <button type="reset">Reset</button>

</form>

<script src="script.js"></script>

</body>

</html>
```

script.js

```
function validateForm() {

    var name = document.getElementById("name").value;
    var email = document.getElementById("email").value;
    var phone = document.getElementById("phone").value;
    var password = document.getElementById("password").value;

    if (name == "") {
        alert("Name cannot be empty");
        return false;
    }

    if (email == "") {
        alert("Email cannot be empty");
        return false;
    }

    if (phone.length != 10 || isNaN(phone)) {
        alert("Phone number must contain 10 digits");
    }
}
```

```
        return false;
    }

    if (password.length < 6) {
        alert("Password must contain at least 6 characters");
        return false;
    }

    alert("Registration Successful");

    return true;
}

body {
    font-family: Arial;
    background: lightgray;
}

h1 {
    text-align: center;
    color: darkblue;
}

form {
    width: 350px;
    margin: auto;
    padding: 20px;
    background: white;
}

label {
    display: block;
    margin-top: 10px;
}

input[type="text"],
```

```
input[type="email"],
input[type="password"],
select {
    width: 100%;
    padding: 8px;
    margin-top: 5px;
    box-sizing: border-box;
}
```

```
button {
    padding: 8px 15px;
    margin-top: 15px;
    background: darkblue;
    color: white;
    border: none;
}
```

PRGM:08

server.js

```
const express = require("express");
const { MongoClient } = require("mongodb");
require("dotenv").config();

const app = express();

app.use(express.json());

const client = new MongoClient(process.env.MONGODB_URI);

async function startServer() {

    await client.connect();

    console.log("Connected to MongoDB");

    const db = client.db("StudentDB");
    const students = db.collection("students");

    // INSERT
    app.post("/students", async (req, res) => {

        const result = await students.insertOne(req.body);

        res.json(result);
    });

    // GET
    app.get("/students", async (req, res) => {

        const result = await students.find().toArray();
```

```

    res.json(result);
  });

  // UPDATE
  app.put("/students/:name", async (req, res) => {

    const result = await students.updateOne(
      { name: req.params.name },
      { $set: req.body }
    );

    res.json(result);
  });

  // DELETE
  app.delete("/students/:name", async (req, res) => {

    const result = await students.deleteOne({
      name: req.params.name
    });

    res.json(result);
  });

  app.listen(3000, () => {
    console.log("Server running on port 3000");
  });
}

startServer();

```

```
npm init -y
```

```
npm install express mongodb dotenv
```

```
curl -X POST http://localhost:3000/students -H "Content-Type: application/json" -d  
"{\"name\":\"Anu\", \"department\":\"Electronics\", \"age\":22}"
```

```
curl http://localhost:3000/students
```

```
curl -X PUT http://localhost:3000/students/Anu -H "Content-Type: application/json" -d  
"{\"age\":23}"
```

```
curl -X DELETE http://localhost:3000/students/Anu
```